



Polymeric material for anti-biofouling

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ABSTRACT

Copolymerization of methyl methacrylate (MMA) and acrylate terminated poly(ethylene oxide-co-ethylene carbonate) (PEOC) macromonomer yields graft copolymer with PEOC side chains. By use of quartz crystal microbalance with dissipation (QCM-D), we have investigated the adsorption of proteins on the surface constructed by the copolymer in seawater. It shows the polymeric surface exhibits protein resistance depending on the content of the side chains. Marine field tests show the protein resistant surface can effectively inhibit marine biofouling for more than 12 weeks.

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1. Introduction

Marine biofouling arises from the undesired accumulation and growth of marine organisms such as microorganisms, barnacles and seaweeds on submerged surfaces. It has adverse effects on shipping and other marine associated activities [1–3]. For example, biofouling on ships can increase the hydrodynamic drag and thereby increases the fuel consumption. So far, the most effective approach for anti-biofouling is to use biocidal coatings containing tributyltin (TBT) formulations. Since the global ban on the use of TBT, most antifouling coatings are based on cuprous oxide in combination with booster biocides, which are also ecologically harmful [4,5]. Recent efforts have been toward environment-friendly antifouling materials including low surface energy elastomers [6–8], bioinspired engineered topographies [9,10], amphiphilic nanostructured coating [11–15], and enzymes [16,17]. On the other hand, some antifoulants made of natural products were also reported [18,19].

It is generally established that biofouling begins with the formation of a biofilm due to the adsorption of microorganism consisting of protein or glycoprotein onto underwater surface [20,21]. Thus, protein resistant surfaces are potentially used to restrain marine biofouling. It is reported that poly(ethylene glycol) (PEG) or oligo(ethylene glycol) (OEG) and zwitterionic polymers, which exhibit protein resistance, can inhibit the settlement and adhesion

of marine algae [22–26]. Yet, only a few of the materials have been examined in natural seawater by field tests [3,24].

In the present study, we have synthesized a set of copolymers of methyl methacrylate (MMA) and acrylate terminated poly(ethylene oxide-co-ethylene carbonate) (PEOC) macromonomer, where PEOC side chains contains protein resistant ethylene oxide (EO) units and degradable carbonate linkage. Meanwhile, the incorporation of poly(methyl methacrylate) (PMMA) can improve the coating ability and mechanical properties of the materials. By use of quartz crystal microbalance with dissipation (QCM-D), we have investigated the adsorption of fibrinogen, bovine serum albumin as well as lysozyme on the surface constructed by the copolymer in natural seawater. The anti-biofouling of the copolymers is also examined by marine field tests. Our aim is to develop non-toxic, effective and durable antifouling material.

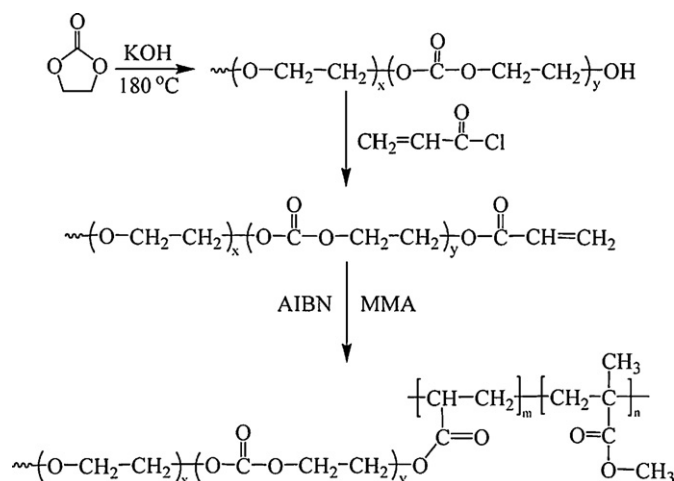
2. Materials and methods

2.1. Materials

Ethylene carbonate (EC), acryloyl chloride (AC), methyl methacrylate (MMA), 2,2-azobisisobutyronitrile (AIBN), potassium hydroxide (KOH) were purchased from Aladdin. EC was dried over CaH₂ and distilled under reduced pressure before use. KOH was dried under vacuum at 110 °C prior to use. MMA was passed through an alkaline Al₂O₃ column to remove the inhibitor before use. AIBN was used after recrystallization twice from methanol. Fibrinogen (fraction I from human plasma, *M_w* = 340 kDa, *pI* = 5.5), lysozyme via chicken egg white (*M_w* = 14.7 kDa, *pI* = 11.1) and BSA

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Scheme 1. Synthesis of PMMA-g-PEOC copolymers.

($M_w = 68$ kDa, $PI = 4.8$) from Merck Chemicals were used as received. Natural seawater was obtained from the site at the inner Xiamen bay ($24^{\circ}45'N$, $118^{\circ}07'E$) in China and filtered before use. Each protein solution (1.0 mg/mL) was prepared by dissolving the protein in natural seawater. Other reagents were used as received.

2.2. Synthesis of the macromonomer

Acrylate terminated poly(ethylene oxide-co-ethylene carbonate) was prepared by ring-opening polymerization of EC at $180^{\circ}C$ using KOH as initiator, followed by the reaction of the terminal hydroxyl groups with acryloyl chloride (Scheme 1). The details can be found elsewhere [27–29]. Typically, EC (88.06 g, 1.0 mol) and KOH (0.056 g, 0.1 mol% based on EC) were introduced into a three-neck 250 ml round-bottomed flask with a magnetic stirrer, a gas inlet/outlet and a condenser. The polymerization was carried out at $180^{\circ}C$ for 9 h under a nitrogen atmosphere. The product was precipitated into ethyl ether and dried under vacuum. 1H NMR (400 MHz, $CDCl_3$, ppm): 4.27 (CH_2CH_2OCOO), 3.71 (CH_2CH_2OCOO), 3.63 (CH_2CH_2O). FTIR: 3400 cm^{-1} (OH), 2875 cm^{-1} (CH_2), 1745 cm^{-1} ($C=O$), 1265 cm^{-1} ($C-O$, the carbonate in EC), 1107 cm^{-1} ($C-O$, the ether in EO).

50 g of the above polymer and 7.6 mL of pyridine were dissolved in 100 mL of THF and 8.0 mL acryloyl chloride was slowly dropped into the mixture at $0^{\circ}C$. The mixture was further stirred at room temperature for 24 h. The reaction solution was filtered and passed through an alkaline Al_2O_3 column to remove the pyridine salt. The product was purified and fractionated by repeated precipitation in ethyl ether from a methanol solution and dried under vacuum. 1H NMR (400 MHz, $CDCl_3$, ppm): 5.8–6.5 ($CH_2=CH$), 4.27 (CH_2CH_2OCOO), 3.71 (CH_2CH_2OCOO), 3.63 (CH_2CH_2O). FTIR: 2875 cm^{-1} (CH_2), 1745 cm^{-1} ($C=O$), 1265 cm^{-1} ($C-O$, the carbonate in EC), 1107 cm^{-1} ($C-O$, the ether in EO). The number average molecular weight (M_n) of the macromonomer determined by 1H NMR was 5300 g/mol and the molecular weight distribution index (M_w/M_n) estimated by gel permeation chromatography (GPC) was 1.32. The molar fraction of EC units estimated by 1H NMR based on the integrals of EC and EO signals was $\sim 20\text{ mol}\%$.

2.3. Synthesis of PMMA-g-PEOC copolymer

The copolymers were synthesized by free radical polymerization (Scheme 1). A certain amount of MMA, PEOC macromonomer, AIBN and THF were added into a three-neck flask equipped with an agitator, a reflux condenser and a gas inlet/outlet. After nitrogen-purged for 30 min, polymerization was carried out at $70^{\circ}C$ for

24 h. The product was precipitated in ethyl ether and washed thoroughly with water and methanol to remove the unreacted macromonomer. The copolymer is designed as PMMA-g-PEOC_x, where x is the weight percentage of PEOC determined by 1H NMR. The characterization data of the PMMA-g-PEOC are shown in Table 1.

2.4. Characterization

2.4.1. Proton nuclear magnetic resonance spectroscopy (1H NMR)

All 1H NMR spectra were recorded on a Bruker AV400 NMR spectrometer using $CDCl_3$ as solvent and tetramethylsilane (TMS) as internal standard.

2.4.2. Fourier transform infrared spectroscopy (FTIR)

FTIR spectra were recorded on a Bruker VECTOR-22 IR spectrometer. The spectra were collected at 64 scans with a spectral resolution of 4 cm^{-1} by the KBr disk method.

2.4.3. Gel permeation chromatography (GPC)

The number and weight-average molecular weights (M_n and M_w) and molecular weight distribution index (M_w/M_n) were measured at $35^{\circ}C$ on a Waters 1515 using a series of monodisperse polystyrenes as standards and THF as the fluent with a flow rate of 1.0 mL/min.

2.4.4. Quartz crystal microbalance with dissipation (QCM-D)

The QCM-D measurements were performed on E1 system with 5 MHz AT-cut quartz crystal (Q-Sense AB). The details can be found elsewhere [30–33]. Briefly, QCM-D simultaneously monitors changes in resonance frequency (Δf) and dissipation (ΔD) in real time. Δf is related to the changes in the mass attached to the oscillating sensor surface, whereas ΔD is to the viscoelasticity of the adsorbed layer [34,35].

For the protein adsorption measurement, natural seawater was used as the reference and the changes in the frequency (Δf) and dissipation (ΔD) give the information about the protein adsorption and structural change of the protein layer. The decrease in the oscillation frequency corresponds to the mass of protein adsorbed on the copolymer surface. All the presented data were from the third overtone ($n = 3$). Δf and ΔD values from the fundamental were discarded because they were usually noise due to insufficient energy trapping [36]. All the experiments were carried out at $25^{\circ}C$ and each was repeated three times. The relative standard deviations for Δf and ΔD were less than 5%.

2.5. Marine field tests

The field tests were performed at the ship coating testing site located at the inner Xiamen bay ($24^{\circ}45'N$, $118^{\circ}07'E$) in China, where the fouling organisms are rich and the marine biofouling is heavy. The fouling organisms attached include hard (barnacles and oysters) and soft foulings (algae, seaweeds, sponges, etc.). During the time period from April until October, the coated panels will experience heavy barnacle fouling pressure due to the warm water environment and the stable salinity. The tested coatings can be readily examined because coating failures can be observed within 4 weeks from immersion due to presence of juvenile barnacles on the surface.

The glass fiber reinforced epoxy resin was used as the panel ($300\text{ mm} \times 250\text{ mm} \times 3\text{ mm}$), which was rubbed with a sandpaper. The coating was applied on each panel to form a thin film and was dried at room temperature for two days. For each coating, three identical panels were prepared and attached to a frame so that they were submerged in different depths in the range 0.2–2.0 m.

Table 1
Characterization data of PMMA-g-PEOC.

Sample	f (wt%) ^a	M_n (g/mol) ^b	M_w (g/mol) ^b	M_w/M_n^b
PMMA-g-PEOC7	6.9	1.19×10^5	2.36×10^5	1.98
PMMA-g-PEOC30	30.4	1.00×10^5	2.05×10^5	2.05
PMMA-g-PEOC45	44.5	0.93×10^5	1.79×10^5	1.93
PMMA-g-PEOC63	63.2	0.85×10^5	1.56×10^5	1.84

^a PEOC content determined by ¹H NMR.

^b Determined by GPC.

The samples were lowered into seawater from a stationary experimental raft that was positioned in the inner Xiamen bay with weak water currents (less than 2 m/s) on May 7, 2011. After a certain period of time, the panels were taken out of the sea and gently washed with seawater and photographed, and then they were immediately placed in the sea to continue the test. Panels coated with PMMA-g-PEOC were tested, where the control panel consisted of the original glass fiber reinforced epoxy resin. Since polyurethane (PEG-PU50) with poly(ethylene glycol) as the soft segment exhibit excellent protein resistance [37], for comparison, a set of panels coated with PEG-PU50 was also examined, where the content of the hard segments is 50 wt%.

3. Results and discussion

Since biofouling formation involves the attachment of protein or glycoprotein, we first examined the protein resistance of the surface constructed by PMMA-g-PEOC copolymer. Fig. 1 shows the time dependence of frequency shift (Δf) and energy dissipation shift (ΔD) for adsorption of fibrinogen on different PMMA-g-PEOC surfaces. Fibrinogen is a large blood plasma protein that can strongly adsorb onto hydrophobic surfaces. For PMMA-g-PEOC7, after fibrinogen is introduced, Δf sharply decreases and then gradually levels off. Upon rinsing with seawater, Δf still shows a marked decrease in comparison with the baseline for seawater. It is known that the increase in mass on the sensor surface causes the frequency to decrease. The decrease in Δf indicates the adsorption of fibrinogen on the surfaces. On the other hand, it is known that ΔD increases with the thickness of the added layer. The marked increase in ΔD relative to the baseline further indicates adsorption of fibrinogen. However, as PEOC content increases, the shifts in Δf and ΔD become smaller, indicating the protein adsorption

decreases. When PEOC content is above 45 wt%, Δf and ΔD return to the baselines after rinsing with seawater, indicating no fibrinogen is adsorbed on the surface. Namely, the PMMA-g-PEOC surface exhibits excellent protein resistance.

Fig. 2 shows the time dependence of frequency shift (Δf) and energy dissipation shift (ΔD) for adsorption of BSA on different PMMA-g-PEOC surfaces. BSA has a smaller size than fibrinogen. Similar to the case of fibrinogen, the Δf and ΔD shifts gradually decrease as PEOC content increases, indicating protein resistance increases. When PEOC content is above 45 wt%, PMMA-g-PEOC can resist the BSA adsorption.

Fig. 3 shows the time dependence of frequency shift (Δf) and energy dissipation shift (ΔD) for the adsorption of lysozyme on different PMMA-g-PEOC surfaces. Lysozyme is the smallest in size among the three proteins, and it is positively charged under the condition (natural seawater, pH $\sim$ 8.2). For PMMA-g-PEOC copolymers, the shifts in Δf and ΔD become smaller as PEOC content increases, indicating the increase in protein resistance. When PEOC content is above 45 wt%, the slight shifts in Δf and ΔD indicate the PMMA-g-PEOC surface can resist the adsorption of lysozyme.

The above studies demonstrate the copolymers with PEOC side chains can resist the nonspecific adsorption of proteins with different sizes and charging levels when PEOC content is high enough. It is known that PEG or poly(ethylene oxide) (PEO) based materials exhibit good protein resistance due to the hydration of the polymers [38–42]. The EO units which make the formation of a tightly bound hydration layer around PEOC chains possible are responsible for the protein resistance of the surface constructed by PMMA-g-PEOC copolymer.

We also examined the anti-biofouling of PMMA-g-PEOC by marine field tests. Fig. 4 shows the marine biofouling of control

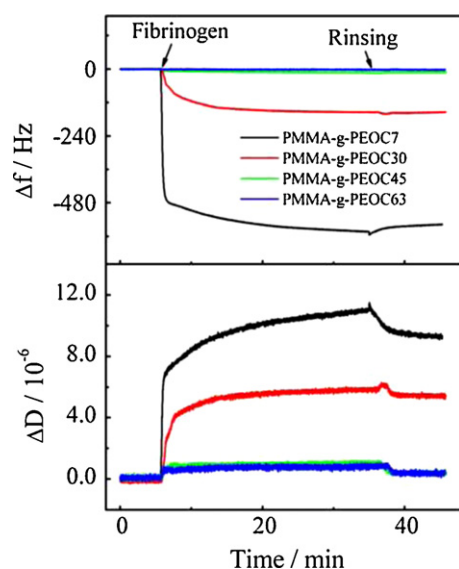


Fig. 1. Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for adsorption of fibrinogen on PMMA-g-PEOC surface in natural seawater at 25 °C.

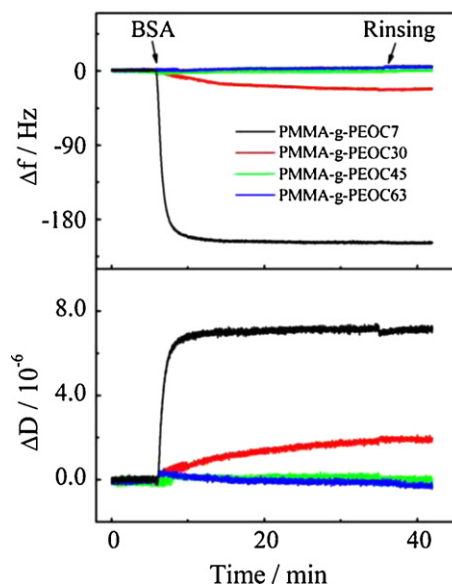


Fig. 2. Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of BSA on PMMA-g-PEOC surface in natural seawater at 25 °C.

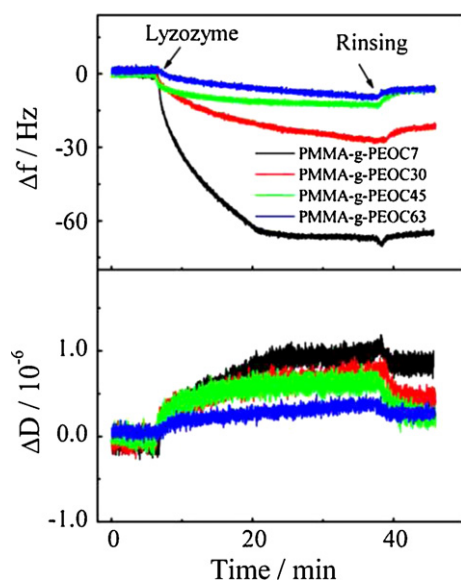


Fig. 3. Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of lysozyme on PMMA-g-PEOC surface in natural seawater at 25 °C.

panels whose surfaces constructed by epoxy resin. As expected, juvenile barnacles already grew on the surface in 4 weeks, and the test panels were fully encrusted by three months of exposure in marine environments. The same things happened on the panels coated with PMMA-g-PEOC when PEOC content was less than 45 wt%. Therefore, the surface with a poor protein resistance does not have anti-biofouling ability.

Fig. 5 shows the biofouling on the panels coated with PMMA-g-PEOC63 whose side chains are protein resistant. We can see that the panels coated with PMMA-g-PEOC63 remain clean within 8 weeks and the surfaces still have good anti-biofouling after 12 weeks. The present results from laboratory experiments and marine field tests in seawater demonstrate that protein resistant PMMA-g-PEOC can effectively inhibit marine biofouling. It is known that when a clean surface is immersed in natural seawater, organisms such as protein or glycoprotein would be quickly adsorbed on the surface and form a conditioning film [21]. Clearly, the PMMA-g-PEOC63 surface can inhibit the biofilm formation and shed fouling organisms. Note that protein resistance is not equal to anti-biofouling [3]. In other words, not all the protein resistant materials can resist marine biofouling in marine environments. We have reported that the surface

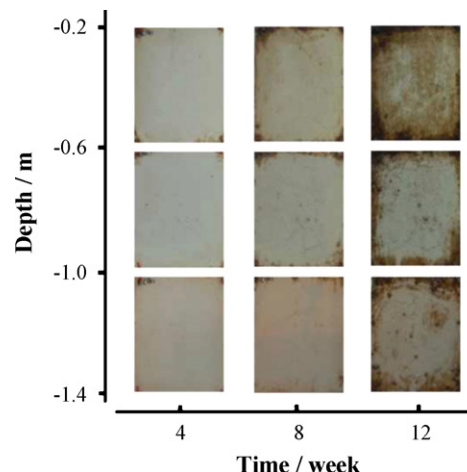


Fig. 5. Images of panels coated with PMMA-g-PEOC63 after immersion in seawater (May–August 2011, Xiamen, China) as a function of time.

constructed by the PEG-PU50 exhibits zero adsorption of proteins [37], where the EO content (~ 50 wt%) is close to that in PMMA-g-PEOC63. The field tests demonstrate that the surface is clean in 4 weeks. However, marine organisms already locate on the surfaces within 8 weeks and are heavily fouled after 12 weeks (Fig. 6). Therefore, protein resistance is not the only reason for anti-biofouling of the present material.

It is reported that PEOC with carbonate linkage susceptible to hydrolysis can degrade in ethanol in the presence of potassium hydroxide [27]. We also managed to investigate the degradation of the graft copolymer in seawater within two days by use of quartz crystal microbalance [43]. Our studies demonstrate that PEOC side chains are degradable in seawater. Note that the marine environment is complex, and the degradation is often accompanied by the attachment of sea mud and organisms on the copolymer surface, so the degradation kinetics of PMMA-g-PEOC copolymer under real conditions in the sea is difficult to detect. To develop a method about in situ detection of the degradation for a long time is a subject of our future work. Anyhow, PMMA-g-PEOC copolymer exhibits both protein resistance and degradation. On the other hand, since the molar fraction of EO units in the copolymer is as high as ~ 80 mol%, they should be relatively uniformly distributed at any thickness, and the protein resistance is expected to vary slightly after a partial degradation. Therefore, the copolymer is expected to form a self-renewed surface with persistent protein resistance,

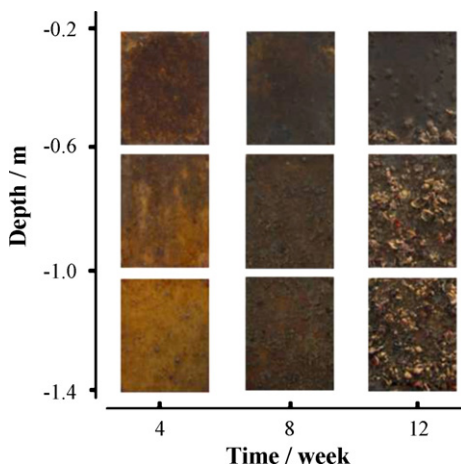


Fig. 4. Images of control panel without coating after immersion in seawater (May–August 2011, Xiamen, China) as a function of time.

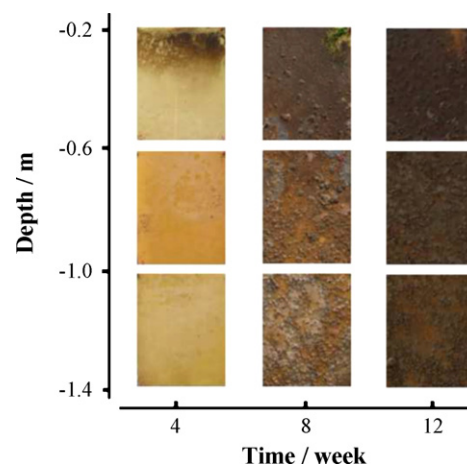


Fig. 6. Images of panels coated with PEG-PU50 after immersion in seawater (May–August 2011, Xiamen, China) as a function of time.

which could polish the attached marine organisms. Actually, the biofouling in marine environments is very complex. Besides marine organisms such as bacteria, spores, diatoms and barnacles, some inorganics (e.g., sea slime) can also accumulate on a surface forming a thin film for biological species to accommodate. It is difficult for a surface with only protein resistance factor to hold long-term anti-biofouling ability because the surface is “dead” and inorganics can adhere to it. The degradable surface with protein resistance in present study is dynamic, so that either living organisms or inorganics can not locate and colonize on it. The combination of protein resistance and hydrolytic degradation leads to the high anti-biofouling performance in marine environments.

4. Conclusion

In conclusion, we have developed copolymers with poly(ethylene oxide-co-ethylene carbonate) side chains. The copolymers can effectively resist nonspecific protein adsorption in natural seawater when the content of side chains is high enough. The marine field tests show that the copolymer has good anti-biofouling ability in marine environments. The present study clearly demonstrates that protein resistance and hydrolytic degradation should be considered when we design materials for marine anti-biofouling.

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